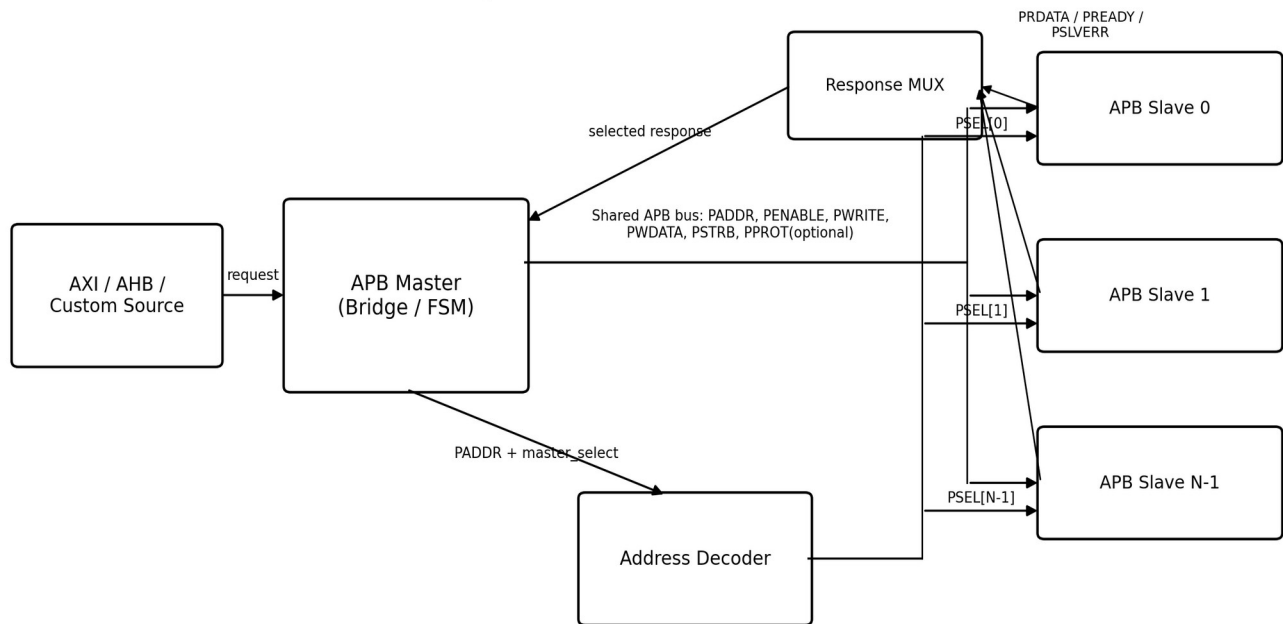


APB4 TOP-LEVEL ARCHITECTURE

One APB4 Master connected to N APB Slaves

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APB4 Top-Level Architecture with N Slaves



Basic idea: I use one APB Master, one Address Decoder, and many APB Slaves. The control/data bus can be shared to all slaves. The decoder checks PADDR and turns on only one PSELx.

1. Basic Architecture

For this design, the APB Master can be a bridge from AXI/AHB, or it can be a custom controller. The Master creates one APB transfer at a time. The Address Decoder uses the address to select one slave.

- APB Master: creates PADDR, PWRITE, PENABLE, PWDATA and PSTRB.
- Address Decoder: checks PADDR and creates PSEL[NUM_SLAVES-1:0].
- APB Slaves: receive the shared APB bus. Only the selected slave responds.
- Response MUX: sends PRDATA, PREADY and PSLVERR from the selected slave back to the Master.
- PCLK and PRESETn are connected to the full APB system.

Simple connection idea:

Master -> shared PADDR/PWRITE/PENABLE/PWDATA/PSTRB -> all slaves
 PADDR -> Address Decoder -> one PSELx becomes 1
 Selected slave -> PRDATA/PREADY/PSLVERR -> Response MUX -> Master

2. APB Master Port List

Port	Dir.	Width	Basic meaning
PCLK	Input	1	APB clock
PRESETn	Input	1	Active-low reset
req_valid / upstream request	Input	1	Request from AXI/AHB/custom logic
req_addr	Input	ADDR_WIDTH	Address from upstream side
req_write	Input	1	Read or write request
req_wdata	Input	DATA_WIDTH	Write data
req_strb	Input	DATA_WIDTH/8	Write byte enable
PREADY	Input	1	Selected slave is ready
PRDATA	Input	DATA_WIDTH	Read data from selected slave
PSLVERR	Input	1	Error from selected slave
PADDR	Output	ADDR_WIDTH	APB address
PENABLE	Output	1	High in ACCESS state
PWRITE	Output	1	1 = write, 0 = read
PWDATA	Output	DATA_WIDTH	APB write data
PSTRB	Output	DATA_WIDTH/8	One bit for each byte
PPROT	Output	3	Optional APB4 protection signal
master_select	Output	1	Internal transfer select for decoder

3. Address Decoder Port List

Port	Dir.	Width	Basic meaning
PADDR	Input	ADDR_WIDTH	Address to decode
master_select	Input	1	Transfer is active
PSEL	Output	NUM_SLAVES	One select bit for each slave

The decoder should normally create a one-hot result. This means only one PSEL bit is HIGH for one valid address.

4. APB Slave Port List

Port	Dir.	Width	Basic meaning
PCLK	Input	1	APB clock
PRESETn	Input	1	Active-low reset
PADDR	Input	ADDR_WIDTH	Shared APB address
PSEL	Input	1	Select for this slave
PENABLE	Input	1	High in ACCESS state
PWRITE	Input	1	1 = write, 0 = read
PWDATA	Input	DATA_WIDTH	Write data
PSTRB	Input	DATA_WIDTH/8	Byte write enable
PPROT	Input	3	Optional APB4 protection signal
PREADY	Output	1	Finish transfer / add wait state
PRDATA	Output	DATA_WIDTH	Read data
PSLVERR	Output	1	Transfer error

5. Main Parameters

Parameter	Example	Meaning
ADDR_WIDTH	32	Width of PADDR. APB allows up to 32-bit address bus.
DATA_WIDTH	32	Width of PWDATA and PRDATA. APB data width can be 8, 16 or 32 bits.
NUM_SLAVES	4	Number of APB slave blocks connected to the interconnect.
PSTRB_WIDTH	DATA_WIDTH/8	Number of write strobe bits.

Basic SystemVerilog parameter example:

```

module apb4_top #(
    parameter int ADDR_WIDTH = 32,
    parameter int DATA_WIDTH = 32,
    parameter int NUM_SLAVES = 4
);

localparam int PSTRB_WIDTH = DATA_WIDTH / 8;

// PSEL has one bit for each slave
logic [NUM_SLAVES-1:0] PSEL;

endmod

```

6. Why PSTRB = DATA_WIDTH / 8

APB4 uses one PSTRB bit for every 8 bits (1 byte) of PWDATA. So the strobe width is DATA_WIDTH divided by 8.

DATA_WIDTH	PSTRB width
8 bits	1 bit
16 bits	2 bits
32 bits	4 bits

Example: with DATA_WIDTH = 32, PSTRB is 4 bits. PSTRB[0] controls PWDATA[7:0], PSTRB[1] controls PWDATA[15:8], and so on.

7. Simple Address Decode Example

For an easy example, I use four slaves. Each slave has one address area. The exact address map is a design choice, not a fixed APB rule.

Slave	Address range	Decoder output
Slave 0	0x0000_0000 - 0x0000_0FFF	PSEL[0] = 1
Slave 1	0x0000_1000 - 0x0000_1FFF	PSEL[1] = 1
Slave 2	0x0000_2000 - 0x0000_2FFF	PSEL[2] = 1
Slave 3	0x0000_3000 - 0x0000_3FFF	PSEL[3] = 1

Basic decode example:

```
always_comb begin
    PSEL = '0;

    if (master_select) begin
        if (PADDR >= 32'h0000_0000 && PADDR <= 32'h0000_0FFF)
            PSEL[0] = 1'b1;
        else if (PADDR >= 32'h0000_1000 && PADDR <= 32'h0000_1FFF)
            PSEL[1] = 1'b1;
        else if (PADDR >= 32'h0000_2000 && PADDR <= 32'h0000_2FFF)
            PSEL[2] = 1'b1;
        else if (PADDR >= 32'h0000_3000 && PADDR <= 32'h0000_3FFF)
            PSEL[3] = 1'b1;
        end
    end
end
```

8. Response MUX

All slaves can see the common APB signals, but only one slave is selected. Because many slaves have PRDATA/PREADY/PSLVERR outputs, the top-level must choose the response from the selected slave.

```
if (PSEL[0]) begin
    PRDATA = PRDATA_S0;
    PREADY = PREADY_S0;
    PSLVERR = PSLVERR_S0;
end
else if (PSEL[1]) begin
    PRDATA = PRDATA_S1;
    PREADY = PREADY_S1;
    PSLVERR = PSLVERR_S1;
end
// ... continue for other slaves
```

9. My Basic Design Summary

- One Master controls the APB transfer.
- PADDR is decoded to create one PSELx for one slave.
- PADDR, PENABLE, PWRITE, PWDATA and PSTRB can be shared to all slaves.
- Only the selected slave should answer the transfer.
- A response MUX returns PRDATA, PREADY and PSLVERR to the Master.
- PSTRB width is DATA_WIDTH/8 because one strobe controls one byte.

Reference used: ARM IHI 0024E - AMBA APB Protocol Specification, especially Chapter 2, Section 3.2, Chapter 4, Appendix A and Appendix B.